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## Requirement Set: StakeholderNeeds

### Description

#### Attributes

|                    |                                                                       |
|--------------------|-----------------------------------------------------------------------|
| <b>Filepath</b>    | D:\dev\galactic-soup\GalacticSoup\requirements\StakeholderNeeds.sreqx |
| <b>Revision</b>    | 1                                                                     |
| <b>Created by</b>  | sarah                                                                 |
| <b>Created on</b>  | 05-Jul-2026 12:23:21                                                  |
| <b>Modified by</b> | sarah                                                                 |
| <b>Modified on</b> | 05-Jul-2026 12:23:30                                                  |

#### Implementation Status

|              |                    |                  |             |
|--------------|--------------------|------------------|-------------|
| <b>Total</b> | <b>Implemented</b> | <b>Justified</b> | <b>None</b> |
| 15           | 0                  | 0                | 15          |

#### Verification Status

|              |               |                  |               |                   |             |
|--------------|---------------|------------------|---------------|-------------------|-------------|
| <b>Total</b> | <b>Passed</b> | <b>Justified</b> | <b>Failed</b> | <b>Unexecuted</b> | <b>None</b> |
| 15           | 0             | 0                | 0             | 0                 | 15          |

**Change Information**    No change issue detected.

## 1 Soup menu

**Requirement Type** Functional

**ID** SN-GS-001

**Description**

Kitchen shall produce a menu of distinct soup varieties selectable by operators.

**Rationale**

Customers expect variety across the menu.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 1     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-001 Recipe count](#) (⇒Derives )

 [SR-GS-002 Throughput](#) (⇒Derives )

 [SR-GS-028 Preparation zone](#) (⇒Derives )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 2 Small crew

**Requirement Type** Functional

**ID** SN-GS-002

**Description**

Kitchen shall be operable by the small crew of beings on site.

**Rationale**

Only 5 beings staff the facility.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 2     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-003 Automation level](#) (⇒Derives )

 [SR-GS-004 Operator count](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

### 3 Galactic shipping

**Requirement Type** Functional

**ID** SN-GS-003

#### Description

Kitchen shall dispatch finished soup to customers across the galaxy.

#### Rationale

Customers are distributed across many worlds.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 3     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-005 Shipping manifest](#) (⇒Derives )

 [SR-GS-006 Transport loading time](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

### 4 Food safety

**Requirement Type** Functional

**ID** SN-GS-004

#### Description

Kitchen shall deliver safe, uncontaminated soup at correct serving temperature.

#### Rationale

Safety and quality are non-negotiable.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 4     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

## Links

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-007 Contamination detection](#) (⇒Derives )

 [SR-GS-008 Serving temperature](#) (⇒Derives )

## Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

## Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 5 Transit-durable packaging

**Requirement Type** Functional

**ID** SN-GS-005

### Description

Packaging shall keep soup intact during long-distance interstellar transit.

### Rationale

Shipments take weeks to reach far customers.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 5     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

## Links

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-009 Container seal life](#) (⇒Derives )

## Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

## Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 6 Inventory tracking

**Requirement Type** Functional

**ID** SN-GS-006

### Description

Kitchen shall track ingredient stock so the crew knows what to reorder.

### Rationale

Stock-outs halt production; crew needs visibility.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 6     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-010 Inventory accuracy](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 7 Facility budgets

**Requirement Type** Functional

**ID** SN-GS-007

#### Description

Kitchen shall fit within facility mass, volume, power, and cost budgets.

#### Rationale

Deployment site imposes hard resource limits.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 7     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.slreqx](#)

 [SR-GS-011 Mass budget](#) (⇒Derives )

 [SR-GS-012 Power budget](#) (⇒Derives )

 [SR-GS-013 Cost budget](#) (⇒Derives )

 [SR-GS-014 Volume budget](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 8 Gravity range

**Requirement Type** Functional

**ID** SN-GS-008

**Description**

Kitchen shall operate across the range of gravitational environments where customer worlds are located.

**Rationale**

Facility may be deployed on any of a diverse set of worlds.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 8     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.sreqx](#)

 [SR-GS-015 Gravity operating range](#) (⇒Derives )

 [SR-GS-016 Structural 12 g tolerance](#) (⇒Derives )

**Implementation Status**

|              |                    |                  |             |
|--------------|--------------------|------------------|-------------|
| <b>Total</b> | <b>Implemented</b> | <b>Justified</b> | <b>None</b> |
| 1            | 0                  | 0                | 1           |

**Verification Status**

|              |               |                  |               |                   |             |
|--------------|---------------|------------------|---------------|-------------------|-------------|
| <b>Total</b> | <b>Passed</b> | <b>Justified</b> | <b>Failed</b> | <b>Unexecuted</b> | <b>None</b> |
| 1            | 0             | 0                | 0             | 0                 | 1           |

## 9 Delivery fleet support

**Requirement Type** Functional

**ID** SN-GS-009

**Description**

Kitchen shall operate and include a fleet of delivery rockets for transporting finished soup to customers and receiving inbound deliveries.

**Rationale**

Dedicated infrastructure required for frequent launch, landing, loading, and refueling operations.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 9     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.sreqx](#)

 [SR-GS-017 Concurrent rocket support](#) (⇒Derives )

 [SR-GS-018 Rocket turnaround](#) (⇒Derives )



 [SR-GS-019 Rocket refueling](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 10 Ingredient Storage

**Requirement Type** Functional

**ID** SN-GS-010

#### Description

Kitchen shall store a variety of delivered ingredients under appropriate temperature and handling conditions.

#### Rationale

Need safe, flexible storage of both perishable and shelf-stable ingredients.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 10    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.sreqx](#)

 [SR-GS-020 Temperature-controlled storage](#) (⇒Derives )

 [SR-GS-021 Ingredient storage capacity](#) (⇒Derives )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 11 Cooking infrastructure

**Requirement Type** Functional

**ID** SN-GS-011

#### Description

Kitchen shall include cooking infrastructure capable of producing soup at required production rates.

#### Rationale

Throughput targets cannot be met without sufficient cooking capacity matched to demand.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 11    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.slreqx](#)  
 [SR-GS-022 Cooking capacity](#) (⇒Derives )

#### Implementation Status

|              |                    |                  |             |
|--------------|--------------------|------------------|-------------|
| <b>Total</b> | <b>Implemented</b> | <b>Justified</b> | <b>None</b> |
| 1            | 0                  | 0                | 1           |

#### Verification Status

|              |               |                  |               |                   |             |
|--------------|---------------|------------------|---------------|-------------------|-------------|
| <b>Total</b> | <b>Passed</b> | <b>Justified</b> | <b>Failed</b> | <b>Unexecuted</b> | <b>None</b> |
| 1            | 0             | 0                | 0             | 0                 | 1           |

## 12 Internal material transport

**Requirement Type** Functional

**ID** SN-GS-012

#### Description

Kitchen shall transport ingredients and products internally between receiving, storage, cooking, packaging, and delivery areas.

#### Rationale

Required to meet throughput targets.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 12    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

#### Links

**Artifact** [SystemRequirements.slreqx](#)  
 [SR-GS-023 Internal transfer time](#) (⇒Derives )  
 [SR-GS-024 Internal transport automation](#) (⇒Derives )

#### Implementation Status

|              |                    |                  |             |
|--------------|--------------------|------------------|-------------|
| <b>Total</b> | <b>Implemented</b> | <b>Justified</b> | <b>None</b> |
| 1            | 0                  | 0                | 1           |

#### Verification Status

|              |               |                  |               |                   |             |
|--------------|---------------|------------------|---------------|-------------------|-------------|
| <b>Total</b> | <b>Passed</b> | <b>Justified</b> | <b>Failed</b> | <b>Unexecuted</b> | <b>None</b> |
| 1            | 0             | 0                | 0             | 0                 | 1           |

## 13 Operational lifecycle

**Requirement Type** Functional

**ID** SN-GS-013

**Description**

Kitchen shall support startup, steady operation, and shutdown across its deployment lifecycle.

**Rationale**

System must operate safely outside of nominal steady-state conditions.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 13    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.slreqx](#)  
 [SR-GS-025 Startup readiness](#) (⇒Derives )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 14 Fault tolerance

**Requirement Type** Functional

**ID** SN-GS-014

**Description**

Kitchen shall continue operating or degrade safely in the presence of faults.

**Rationale**

Production continuity and safety must be maintained with minimal crew intervention.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 14    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

**Links**

**Artifact** [SystemRequirements.slreqx](#)  
 [SR-GS-026 Fault-induced throughput degradation](#) (⇒Derives )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 15 System coordination

**Requirement Type** Functional

**ID** SN-GS-015

### Description

Kitchen shall coordinate cooking, packaging, storage, and transport activities to meet operational goals.

### Rationale

High automation requires integrated system-level control.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 15    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:22 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:22 |

**Change Information** No change issue detected.

### Links

**Artifact** [SystemRequirements.sreqx](#)

 [SR-GS-027 Production coordination](#) (⇒Derives )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 0           | 0         | 1    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## Requirement Set: SystemRequirements

### Description

### Attributes

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| <b>Filepath</b>    | D:\dev\galactic-soup\GalacticSoup\requirements\SystemRequirements.slreqx |
| <b>Revision</b>    | 1                                                                        |
| <b>Created by</b>  | sarah                                                                    |
| <b>Created on</b>  | 05-Jul-2026 12:23:23                                                     |
| <b>Modified by</b> | sarah                                                                    |
| <b>Modified on</b> | 07-Jul-2026 21:19:43                                                     |

### Implementation Status

|              |                    |                  |             |
|--------------|--------------------|------------------|-------------|
| <b>Total</b> | <b>Implemented</b> | <b>Justified</b> | <b>None</b> |
| 28           | 28                 | 0                | 0           |

### Verification Status

|              |               |                  |               |                   |             |
|--------------|---------------|------------------|---------------|-------------------|-------------|
| <b>Total</b> | <b>Passed</b> | <b>Justified</b> | <b>Failed</b> | <b>Unexecuted</b> | <b>None</b> |
| 28           | 8             | 0                | 0             | 0                 | 20          |

**Change Information**    No change issue detected.

## 1 Recipe count

**Requirement Type** Functional

**ID** SR-GS-001

### Description

System shall cook at least 8 distinct recipes selectable at runtime.

### Rationale

Eight recipes gives menu variety while staying within ingredient storage and recipe-validation budgets.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 1     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:5 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-001 Soup menu](#) (↔Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

 [HyperCook recipe campaign](#) (↔Verified by )

 [EverSimmer recipe campaign](#) (↔Verified by )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 2 Throughput

**Requirement Type** Functional

**ID** SR-GS-002

### Description

System shall sustain total throughput of at least 200 bowls/hour.

### Rationale

200 bowls/hour matches expected peak order rate aggregated across customer worlds.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 2     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

**Links**

**Artifact** [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate\_2.slreqx:7 (←Refined by )

**Artifact** [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:5 (←Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

📄 [SN-GS-001 Soup menu](#) (←Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

📄 [HyperCook nominal](#) (←Verified by )

📄 [EverSimmer nominal](#) (←Verified by )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

### 3 Automation level

**Requirement Type** Functional

**ID** SR-GS-003

**Description**

System shall achieve average automation level of at least 0.8 across components.

**Rationale**

0.8 automation lets the 5-operator crew cover a 24/7 facility with rotation and breaks.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 3     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

**Links**

**Artifact** [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate\_2.slreqx:8 (←Refined by )

**Artifact** [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (←Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

📄 [SN-GS-002 Small crew](#) (←Derived from )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
|-------|-------------|-----------|------|

|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 0 | 0 |
|---|---|---|---|

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 4 Operator count

**Requirement Type** Functional

**ID** SR-GS-004

#### Description

System shall require at most 5 concurrent operators at peak load.

#### Rationale

Facility berthing and life support are sized for a 5-person crew.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 4     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate\_2.slreqx:9 (↔Refined by )

**Artifact** [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

📄 [SN-GS-002 Small crew](#) (↔Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 5 Shipping manifest

**Requirement Type** Functional

**ID** SR-GS-005

#### Description

System shall generate a shipping manifest per batch including destination.

#### Rationale

Manifest is required for customs, transport routing, and recipient confirmation on delivery.

#### Revision Information

|            |   |                 |   |
|------------|---|-----------------|---|
| <b>SID</b> | 5 | <b>Revision</b> | 1 |
|------------|---|-----------------|---|



|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:8 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-003 Galactic shipping](#) (↔Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 6 Transport loading time

**Requirement Type** Functional

**ID** SR-GS-006

#### Description

System shall load packaged soup onto transport within 10 minutes of packaging.

#### Rationale

10-minute window keeps loading pipeline aligned with the 200 bowls/hour production rate.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 6     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:8 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-003 Galactic shipping](#) (↔Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

 [HyperCook transport loading](#) (↔Verified by )

 [LeanBroth transport loading](#) (↔Verified by )

 [EverSimmer transport loading](#) (↔Verified by )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

**7 Contamination detection****Requirement Type** Functional**ID** SR-GS-007**Description**

System shall detect contamination before sealing with at least 99% sensitivity.

**Rationale**

99% sensitivity balances false-positive rework cost against the downstream cost of a contaminated shipment.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 7     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.**Links****Artifact** [GalacticSoupFunctional.slx](#) GalacticSoupFunctional:6 (↔Implemented by )**Artifact** [PhysicalEverSimmer.slx](#) [CellQCSensor](#) (↔Implemented by ) [CellQCSensor](#) (↔Implemented by ) [CellQCSensor](#) (↔Implemented by )**Artifact** [PhysicalHyperCook.slx](#) [InlineQCScanner](#) (↔Implemented by )**Artifact** [PhysicalLeanBroth.slx](#) [QCBench](#) (↔Implemented by )**Artifact** [StakeholderNeeds.slsreqx](#) [SN-GS-004 Food safety](#) (↔Derived from )**Artifact** [GalacticSoupSystemTests.mldatx](#) [HyperCook contamination 2 percent](#) (↔Verified by ) [EverSimmer contamination 2 percent](#) (↔Verified by )**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 8 Serving temperature

**Requirement Type** Functional

**ID** SR-GS-008

**Description**

System shall verify soup temperature in 70-95 C before QC sign-off.

**Rationale**

70-95 C is the safe serving range: above pathogen-kill threshold and below scald risk.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 8     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

**Links**

**Artifact** [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:6 (↔Implemented by )

**Artifact** [PhysicalEverSimmer.slx](#)

☐ [CellQCSensor](#) (↔Implemented by )

☐ [CellQCSensor](#) (↔Implemented by )

☐ [CellQCSensor](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

☐ [InlineQCScanner](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

☐ [QCBench](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

☐ [SN-GS-004 Food safety](#) (↔Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

☐ [EverSimmer nominal](#) (↔Verified by )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 9 Container seal life

**Requirement Type** Functional

**ID** SR-GS-009

**Description**

System shall seal containers rated for 30-day interstellar transit.

### Rationale

30 days covers typical transit time to the farthest customer worlds with margin.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 9     | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:7 (↔Implemented by )

**Artifact** [PhysicalEverSimmer.slx](#)

 [CellPackager](#) (↔Implemented by )

 [CellPackager](#) (↔Implemented by )

 [CellPackager](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

 [HighSpeedPackagingLine](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

 [SemiAutoPackager](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.slsreqx](#)

 [SN-GS-005 Transit-durable packaging](#) (↔Derived from )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 10 Inventory accuracy

**Requirement Type** Functional

**ID** SR-GS-010

### Description

System shall track ingredient inventory with at most 1% stock error.

### Rationale

1% stock error keeps reorder decisions accurate enough to avoid production-halting stock-outs.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 10    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

## Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:3 (↔Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

📄 [SN-GS-006 Inventory tracking](#) (↔Derived from )

## Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

## Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

# 11 Mass budget

Requirement Type Functional

ID SR-GS-011

## Description

Total system mass shall not exceed 15000 kg.

## Rationale

Allocated from the facility deployment mass budget.

## Revision Information

|             |       |             |                      |
|-------------|-------|-------------|----------------------|
| SID         | 11    | Revision    | 1                    |
| Created by  | sarah | Created on  | 05-Jul-2026 12:23:23 |
| Modified by | sarah | Modified on | 05-Jul-2026 12:23:23 |

Change Information No change issue detected.

## Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate\_2.slsreqx:3 (↔Refined by )

Artifact [PhysicalEverSimmer.slx](#)

📄 [PhysicalEverSimmer](#) (↔Implemented by )

Artifact [PhysicalHyperCook.slx](#)

📄 [PhysicalHyperCook](#) (↔Implemented by )

Artifact [PhysicalLeanBroth.slx](#)

📄 [PhysicalLeanBroth](#) (↔Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

📄 [SN-GS-007 Facility budgets](#) (↔Derived from )

## Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

## Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
|-------|--------|-----------|--------|------------|------|

|   |   |   |   |   |   |
|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 1 |
|---|---|---|---|---|---|

## 12 Power budget

**Requirement Type** Functional

**ID** SR-GS-012

### Description

Total system power draw shall not exceed 500 kW.

### Rationale

Allocated from the facility power budget.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 12    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate\_2.slreqx:4 (↔Refined by )

**Artifact** [PhysicalEverSimmer.slx](#)

📦 [PhysicalEverSimmer](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

📦 [PhysicalHyperCook](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

📦 [PhysicalLeanBroth](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

📋 [SN-GS-007 Facility budgets](#) (↔Derived from )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 13 Cost budget

**Requirement Type** Functional

**ID** SR-GS-013

### Description

Total system cost shall not exceed 2000000 credits.

### Rationale

Allocated from the facility capital cost budget.

### Revision Information

|            |    |                 |   |
|------------|----|-----------------|---|
| <b>SID</b> | 13 | <b>Revision</b> | 1 |
|------------|----|-----------------|---|

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupComplianceGate.slx](#)

 GalacticSoupComplianceGate\_2.slreqx:5 (↔Refined by )

**Artifact** [PhysicalEverSimmer.slx](#)

 [PhysicalEverSimmer](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

 [PhysicalHyperCook](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

 [PhysicalLeanBroth](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-007 Facility budgets](#) (↔Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 14 Volume budget

**Requirement Type** Functional

**ID** SR-GS-014

#### Description

Total system volume shall not exceed 400 m<sup>3</sup>.

#### Rationale

Allocated from the facility floor-space and volume budget.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 14    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupComplianceGate.slx](#)

 GalacticSoupComplianceGate\_2.slreqx:6 (↔Refined by )

**Artifact** [PhysicalEverSimmer.slx](#)

 [PhysicalEverSimmer](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

 [PhysicalHyperCook](#) (↔Implemented by )

Artifact [PhysicalLeanBroth.slx](#)

 [PhysicalLeanBroth](#) (←Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-007 Facility budgets](#) (←Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 15 Gravity operating range

Requirement Type Functional

ID SR-GS-015

#### Description

System shall perform all cooking, packaging, and shipping functions nominally across ambient gravity 0.1 g to 12 g.

#### Rationale

0.1-12 g spans the full ambient gravity range across candidate customer worlds.

#### Revision Information

|             |       |             |                      |
|-------------|-------|-------------|----------------------|
| SID         | 15    | Revision    | 1                    |
| Created by  | sarah | Created on  | 05-Jul-2026 12:23:23 |
| Modified by | sarah | Modified on | 05-Jul-2026 12:23:23 |

Change Information No change issue detected.

#### Links

Artifact [GalacticSoupComplianceGate.slx](#)

 [GalacticSoupComplianceGate\\_2.slsreqx:10](#) (←Refined by )

Artifact [GalacticSoupFunctional.slx](#)

 [GalacticSoupFunctional:12](#) (←Implemented by )

Artifact [PhysicalEverSimmer.slx](#)

 [GravityCompMesh](#) (←Implemented by )

Artifact [PhysicalHyperCook.slx](#)

 [GravityCompensatorArray](#) (←Implemented by )

Artifact [PhysicalLeanBroth.slx](#)

 [GravityCompUnit](#) (←Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-008 Gravity range](#) (←Derived from )

Artifact [GalacticSoupSystemTests.mldatx](#)

 [HyperCook at 0.1 g](#) (←Verified by )

 [HyperCook at 12 g](#) (←Verified by )



**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 16 Structural 12 g tolerance

**Requirement Type** Functional**ID** SR-GS-016**Description**

System structure and mounts shall withstand sustained 12 g loading without permanent deformation.

**Rationale**

12 g is the worst-case sustained ambient gravity among customer worlds.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 16    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.**Links****Artifact** [GalacticSoupFunctional.slx](#) GalacticSoupFunctional:12 (↔Implemented by )**Artifact** [PhysicalEverSimmer.slx](#)☐ [GravityCompMesh](#) (↔Implemented by )**Artifact** [PhysicalHyperCook.slx](#)☐ [GravityCompensatorArray](#) (↔Implemented by )**Artifact** [PhysicalLeanBroth.slx](#)☐ [GravityCompUnit](#) (↔Implemented by )**Artifact** [StakeholderNeeds.slreqx](#) [SN-GS-008 Gravity range](#) (↔Derived from )**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 17 Concurrent rocket support

**Requirement Type** Functional**ID** SR-GS-017

### Description

System shall support simultaneous handling of at least 3 delivery rockets for loading or unloading.

### Rationale

Parallel handling prevents outbound shipping from blocking inbound ingredient deliveries.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 17    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:9 (↔Implemented by )

**Artifact** [PhysicalEverSimmer.slx](#)

☐ [TriplePadPort](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

☐ [LaunchPadComplex](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

☐ [TriPadLandingField](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

📄 [SN-GS-009 Delivery fleet support](#) (↔Derived from )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 18 Rocket turnaround

**Requirement Type** Functional

**ID** SR-GS-018

### Description

System shall complete loading or unloading of a delivery rocket within 20 minutes.

### Rationale

Fast turnaround maximizes fleet utilization and maintains alignment with production throughput.

### Revision Information

|                   |       |                   |                      |
|-------------------|-------|-------------------|----------------------|
| <b>SID</b>        | 18    | <b>Revision</b>   | 1                    |
| <b>Created by</b> | sarah | <b>Created on</b> | 05-Jul-2026 12:23:23 |

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |
|--------------------|-------|--------------------|----------------------|

**Change Information** No change issue detected.

#### Links

- Artifact** [GalacticSoupFunctional.slx](#)  
 GalacticSoupFunctional:9 (↔Implemented by )  
**Artifact** [PhysicalEverSimmer.slx](#)  
☐ [AutoCargoLoader](#) (↔Implemented by )  
**Artifact** [PhysicalHyperCook.slx](#)  
☐ [CargoLoaderGantry](#) (↔Implemented by )  
**Artifact** [PhysicalLeanBroth.slx](#)  
☐ [SharedCraneDock](#) (↔Implemented by )  
**Artifact** [StakeholderNeeds.slreqx](#)  
☐ [SN-GS-009 Delivery fleet support](#) (↔Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 19 Rocket refueling

**Requirement Type** Functional

**ID** SR-GS-019

#### Description

System shall provide on-site refueling capability for supported delivery rockets.

#### Rationale

On-site refueling minimizes logistics dependencies and enables rapid redeployment.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 19    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

- Artifact** [GalacticSoupFunctional.slx](#)  
 GalacticSoupFunctional:9 (↔Implemented by )  
**Artifact** [PhysicalEverSimmer.slx](#)  
☐ [AutoRefuelCell](#) (↔Implemented by )  
**Artifact** [PhysicalHyperCook.slx](#)  
☐ [RefuelingStation](#) (↔Implemented by )

Artifact [PhysicalLeanBroth.slx](#)

 [RefuelSkid](#) (←Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-009 Delivery fleet support](#) (←Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 20 Temperature-controlled storage

Requirement Type Functional

ID SR-GS-020

#### Description

System shall provide separate storage zones for cold and room-temperature ingredients.

#### Rationale

Different ingredient classes require different storage conditions to maintain safety and quality.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 20    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

Change Information No change issue detected.

#### Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:2 (←Implemented by )

Artifact [PhysicalEverSimmer.slx](#)

 [DualZoneStore](#) (←Implemented by )

Artifact [PhysicalHyperCook.slx](#)

 [ColdStorageVault](#) (←Implemented by )

Artifact [PhysicalLeanBroth.slx](#)

 [ColdStoreLocker](#) (←Implemented by )

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-010 Ingredient Storage](#) (←Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 21 Ingredient storage capacity

**Requirement Type** Functional

**ID** SR-GS-021

### Description

System shall store at least 72 hours of ingredients at nominal production rate.

### Rationale

72 hours of buffer mitigates delivery delays without exceeding storage mass and volume budgets.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 21    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:2 (↔Implemented by )

**Artifact** [PhysicalEverSimmer.slx](#)

 [DualZoneStore](#) (↔Implemented by )

**Artifact** [PhysicalHyperCook.slx](#)

 [AmbientStorageSilo](#) (↔Implemented by )

**Artifact** [PhysicalLeanBroth.slx](#)

 [DryGoodsRack](#) (↔Implemented by )

**Artifact** [StakeholderNeeds.sreqx](#)

 [SN-GS-010 Ingredient Storage](#) (↔Derived from )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 22 Cooking capacity

**Requirement Type** Functional

**ID** SR-GS-022

### Description

System shall cook soup at a sustained rate sufficient to meet overall throughput requirements.

### Rationale

Cooking must not be the bottleneck in achieving 200 bowls/hour system throughput.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 22    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:5 (←Implemented by )

**Artifact** [StakeholderNeeds.slsreqx](#)

 [SN-GS-011 Cooking infrastructure](#) (←Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

#### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 23 Internal transfer time

**Requirement Type** Functional

**ID** SR-GS-023

#### Description

System shall transport ingredients from storage to cooking at a rate sufficient to sustain nominal cooking throughput.

#### Rationale

Cooking throughput must not be limited by internal ingredient transport.

#### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 23    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

#### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:10 (←Implemented by )

**Artifact** [StakeholderNeeds.slsreqx](#)

 [SN-GS-012 Internal material transport](#) (←Derived from )

#### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 24 Internal transport automation

**Requirement Type** Functional**ID** SR-GS-024**Description**

Internal material transport shall be automated for at least 90% of transfers.

**Rationale**

High automation supports low crew size and reduces operator workload.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 24    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.**Links****Artifact** [GalacticSoupFunctional.slx](#) GalacticSoupFunctional:10 (↔Implemented by )**Artifact** [StakeholderNeeds.slsrx](#) [SN-GS-012 Internal material transport](#) (↔Derived from )**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 25 Startup readiness

**Requirement Type** Functional**ID** SR-GS-025**Description**

System shall reach nominal operating throughput within a defined startup period after activation.

**Rationale**

Ensures the system can transition from inactive to productive operation without extended downtime.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 25    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

**Links**

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:11 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-013 Operational lifecycle](#) (↔Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

 [HyperCook nominal](#) (↔Verified by )

 [EverSimmer nominal](#) (↔Verified by )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 26 Fault-induced throughput degradation

**Requirement Type** Functional

**ID** SR-GS-026

**Description**

System shall prevent uncontrolled termination of production due to single internal component faults.

**Rationale**

Should tolerate individual component failures without complete loss of production.

**Revision Information**

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 26    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

**Links**

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:11 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-014 Fault tolerance](#) (↔Derived from )

**Artifact** [GalacticSoupSystemTests.mldatx](#)

 [EverSimmer worst fault](#) (↔Verified by )

**Implementation Status**

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

**Verification Status**



| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 1      | 0         | 0      | 0          | 0    |

## 27 Production coordination

**Requirement Type** Functional

**ID** SR-GS-027

### Description

System shall coordinate internal operations to satisfy throughput, safety, and logistical constraints concurrently.

### Rationale

Ensures that throughput, safety, and logistics objectives are met simultaneously in a highly automated system.

### Revision Information

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>SID</b>         | 27    | <b>Revision</b>    | 1                    |
| <b>Created by</b>  | sarah | <b>Created on</b>  | 05-Jul-2026 12:23:23 |
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |

**Change Information** No change issue detected.

### Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:11 (←Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-015 System coordination](#) (←Derived from )

### Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

### Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## 28 Preparation zone

**Requirement Type** Functional

**ID** SR-GS-028

### Description

System shall include a separate ingredient preparation zone supporting chopping and weighing of ingredients, physically isolated from storage and cooking zones. The zone shall sustain prep throughput of at least 200 bowls/hour equivalent.

### Rationale

A dedicated prep zone keeps raw/prepped materials separated and sized so it does not bottleneck the cooking line.

### Revision Information

|                   |       |                   |                      |
|-------------------|-------|-------------------|----------------------|
| <b>SID</b>        | 28    | <b>Revision</b>   | 1                    |
| <b>Created by</b> | sarah | <b>Created on</b> | 05-Jul-2026 12:23:23 |

|                    |       |                    |                      |
|--------------------|-------|--------------------|----------------------|
| <b>Modified by</b> | sarah | <b>Modified on</b> | 05-Jul-2026 12:23:23 |
|--------------------|-------|--------------------|----------------------|

**Change Information** No change issue detected.

## Links

**Artifact** [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:4 (↔Implemented by )

**Artifact** [StakeholderNeeds.slreqx](#)

 [SN-GS-001 Soup menu](#) (↔Derived from )

## Implementation Status

| Total | Implemented | Justified | None |
|-------|-------------|-----------|------|
| 1     | 1           | 0         | 0    |

## Verification Status

| Total | Passed | Justified | Failed | Unexecuted | None |
|-------|--------|-----------|--------|------------|------|
| 1     | 0      | 0         | 0      | 0          | 1    |

## Appendix

### Artifact List

Requirement Set files:

| # | Name                      | Folder                                         | Revision |
|---|---------------------------|------------------------------------------------|----------|
| 1 | SystemRequirements.slreqx | D:\dev\galactic-soup\GalacticSoup\requirements | 1        |
| 2 | StakeholderNeeds.slrreqx  | D:\dev\galactic-soup\GalacticSoup\requirements | 1        |

Simulink models:

| # | Name                           | Folder                                         | Version   |
|---|--------------------------------|------------------------------------------------|-----------|
| 1 | GalacticSoupFunctional.slx     | D:\dev\galactic-soup\GalacticSoup\architecture | Unloaded. |
| 2 | GalacticSoupComplianceGate.slx | D:\dev\galactic-soup\GalacticSoup\architecture | Unloaded. |
| 3 | PhysicalEverSimmer.slx         | D:\dev\galactic-soup\GalacticSoup\architecture | 1.20      |
| 4 | PhysicalHyperCook.slx          | D:\dev\galactic-soup\GalacticSoup\architecture | 1.17      |
| 5 | PhysicalLeanBroth.slx          | D:\dev\galactic-soup\GalacticSoup\architecture | 1.17      |

Simulink Test files:

| # | Name                           | Folder                                  | File timestamp       |
|---|--------------------------------|-----------------------------------------|----------------------|
| 1 | GalacticSoupSystemTests.mldatx | D:\dev\galactic-soup\GalacticSoup\tests | 07-Jul-2026 21:19:41 |